

CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 01: From Python to Java

Name (Print): _____ Section: _____

1. **Python → Java** Compare the following programs.

Python

```
diamonds = 3
player = "Steve"

if diamonds >= 3:
    print("Craft a pickaxe")
```

Java

```
int diamonds = 3;
String player = "Steve";

if (diamonds >= 3) {
    System.out.println("Craft a pickaxe");
}
```

Identify **three differences** you notice between the Python and Java code.

- (a) _____
 (b) _____
 (c) _____

2. **Data Types.** Match each Java declaration with the kind of value it stores.

- | | |
|----------------------|-----------|
| (a) blocks = 64; | A double |
| (b) health = 18.5; | B int |
| (c) hasSword = true; | C char |
| (d) rank = 'A'; | D boolean |
| (e) player = "Alex"; | E String |

3. Which one above is **not a primitive type**? _____

4. **Variables and Operators.** Determine the final value of each variable.

```
1 int a = 5;
2 int b = 3;
3 int c = 4;
4 c = a + b;
5 c = ?
```

5. **Strings.** Consider `String player = "Steve";`

```
1 player.length()
2 player.charAt(0)
3 player.substring(0, 3)
4 "Hi " + player
5 \end{enumerate}
```

Write the result of each expression.

(a) _____

(b) _____

(c) _____

(d) _____

6. **Challenge: + Operator.** Predict the Output:

(a) $2 + 3 =$

(b) `'2' + '3' =`

(c) `"2" + "3" =`

(d) `"2" + '3' =`

7. **Conditionals**

```
1 int g(int a) {
2     if (a < 9) {
3         return 9;
4     }
5     if (a < 7) {
6         return 7;
7     }
8     if (a < 4) {
9         return 4;
10    }
11    return 0;
12 }
```

What value is returned by:

(a) `g(5) =` _____

(b) `g(10) =` _____

8. **For Loops**

```
1 int blocks = 0;
2 for (int i = 0; i < 4; i++) {
3     blocks += 2;
4 }
```

Final value of `blocks`: _____

9. **Before You Leave**

What is one thing from this topic that you found interesting, OR what is one question you are now curious to learn more about?